Module B3 — Physics in Action

3.1 Imaging and Signalling

Lenses and Image Formation

Converging (convex) lens: A lens that causes parallel rays of light to converge to a focal point. It is thicker in the middle than at the edges.

Diverging (concave) lens: A lens that causes parallel rays of light to diverge as though they came from a focal point on the same side as the incoming light. It is thinner in the middle.

The behaviour of lenses is described by the thin lens equation:

$$1/v = 1/u + 1/f$$

where u is the object distance (from lens to object), v is the image distance (from lens to image), and f is the focal length of the lens. All distances are measured from the centre of the lens.

Focal length (f): The distance from the centre of the lens to the principal focus, where parallel rays converge (for a converging lens) or appear to diverge from (for a diverging lens).

Power of a lens (P): The reciprocal of the focal length, measured in dioptres (D). P = 1/f, where f is in metres. A more powerful lens has a shorter focal length and bends light more strongly.

$$\text{Power: } P = 1/f \quad (P \text{ in dioptres, } f \text{ in metres})$$

For a converging lens, f is positive and P is positive. For a diverging lens, f is negative and P is negative. When lenses are combined in contact, their powers add: $P_{total} = P_1 + P_2$.

Sign Convention

OCR B uses the real-is-positive convention:

- Real object distances (u) are positive (object on the same side as incoming light)
- Real image distances (v) are positive (image on the opposite side from the object)
- Virtual image distances are negative
- Converging lenses have positive focal length; diverging lenses have negative focal length

Magnification

$$\text{Linear magnification: } m = \text{image height} / \text{object height} = v / u$$

If |m| > 1, the image is magnified. If |m| < 1, the image is diminished. A negative magnification indicates the image is inverted.

Exam Tip: When using the lens equation, always be careful with signs. A common mistake is to forget that for a diverging lens, f is negative. Substitute values with their correct signs and check that the sign of your answer for v makes physical sense.

Ray Diagrams

Three standard rays can be drawn for a converging lens:

1. A ray parallel to the principal axis refracts through the focal point on the other side
2. A ray through the centre of the lens passes straight through undeviated
3. A ray through the focal point on the object side refracts to emerge parallel to the principal axis

The image is formed where at least two of these rays intersect. For objects beyond 2f, the image is real, inverted, and diminished. At 2f, it is the same size. Between f and 2f, it is real, inverted, and magnified. Inside f, the image is virtual, upright, and magnified.

3.2 Refraction

Refraction: The change in direction of a wave when it passes from one medium to another due to a change in wave speed. The frequency remains constant while the wavelength and speed change.

$$\text{Snell's law: } n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Refractive index (n): A measure of how much a medium slows down light. n = c / v, where c is the speed of light in a vacuum and v is the speed of light in the medium. The refractive index of a vacuum is 1; for air it is approximately 1.00; for glass it is typically 1.5; for water it is about 1.33.

Total Internal Reflection

When light travels from a more optically dense medium to a less dense one ($n_1 > n_2$), there is a critical angle beyond which all light is reflected back into the denser medium.

$$\text{Critical angle: } \sin \theta_c = n_2 / n_1$$

For angles of incidence greater than the critical angle, total internal reflection occurs. This is the principle behind optical fibres, which transmit light signals over long distances with minimal loss.

Optical Fibres

Optical fibres use total internal reflection to guide light along a thin glass or plastic core. They consist of a core with a higher refractive index surrounded by cladding with a lower refractive index. Applications include telecommunications and medical endoscopy.

Signal degradation in optical fibres occurs through:

- **Absorption:** The fibre material absorbs some light energy, converting it to heat
- **Dispersion (pulse broadening):** Different wavelengths travel at slightly different speeds (material dispersion) and rays entering at different angles take different path lengths (modal dispersion), causing pulses to spread out and overlap

3.3 Diffraction

Diffraction: The spreading of waves when they pass through a gap or around an obstacle. Diffraction is most significant when the size of the gap or obstacle is comparable to the wavelength of the wave.

Single Slit Diffraction

When monochromatic light passes through a single narrow slit, it produces a diffraction pattern consisting of a bright central maximum flanked by alternating dark and bright fringes of decreasing intensity. The central maximum is twice the width of the subsidiary maxima.

Diffraction Gratings

A diffraction grating consists of many equally spaced parallel slits. When monochromatic light passes through, it produces sharp, bright maxima at specific angles determined by the grating equation:

$$d \sin \theta = n\lambda$$

where d is the slit spacing (distance between centres of adjacent slits), θ is the angle of diffraction, n is the order number (0, 1, 2, ...), and λ is the wavelength. The slit spacing d = 1/N where N is the number of slits per metre.

Exam Tip: The maximum number of orders visible is found by setting sin θ = 1 (the maximum possible value), giving $n_{max} = d/\lambda$, rounded down to the nearest integer. Remember there are orders on both sides of the central maximum.

3.4 Digital Signals and Communication

Analogue vs Digital Signals

Analogue signal: A signal that varies continuously and can take any value within a range.

Digital signal: A signal that has only a limited number of discrete values, typically two levels (0 and 1, or high and low).

Digital transmission has several advantages over analogue:

- **Noise immunity:** Digital signals can be regenerated perfectly at repeater stations, removing accumulated noise. Analogue signals can only be amplified, which amplifies the noise too.
- **Multiplexing:** Multiple digital signals can be combined and transmitted simultaneously through the same channel using time-division multiplexing
- **Encryption:** Digital data can be easily encrypted for security
- **Processing:** Digital signals can be processed, stored, and manipulated by computers

Sampling and Bit Depth

To convert an analogue signal to digital, the signal is sampled at regular intervals and each sample is assigned a numerical value (quantisation). The quality of the digital representation depends on:

- **Sampling rate:** How frequently the signal is sampled (in Hz). Must be at least twice the maximum frequency of the signal (Nyquist criterion) to accurately represent it
- **Bit depth (resolution):** The number of bits used to represent each sample. More bits allow more possible values and a finer representation of the original signal

$$\text{Number of possible levels} = 2^{n} \quad (\text{where } n = \text{number of bits})$$

3.5 Materials

Stress, Strain, and Young's Modulus

Stress (σ): The force applied per unit cross-sectional area. Measured in pascals (Pa). $\sigma = F/A$

Strain (ε): The extension per unit original length. Strain has no unit (it is a ratio). $\varepsilon = \Delta L/L$

Young's modulus (E): The ratio of stress to strain in the region where the material obeys Hooke's law. Measured in Pa. $E = \sigma/\varepsilon$

$$\text{Stress: } \sigma = F / A$$

$$\text{Strain: } \varepsilon = \Delta L / L$$

$$\text{Young's modulus: } E = \sigma / \varepsilon = FL / A\Delta L$$

Young's modulus is a property of the material, not the object. A high Young's modulus means the material is stiff and resists deformation. Steel has a Young's modulus of about 200 GPa; rubber is about 0.01 GPa.

Stress-Strain Graphs

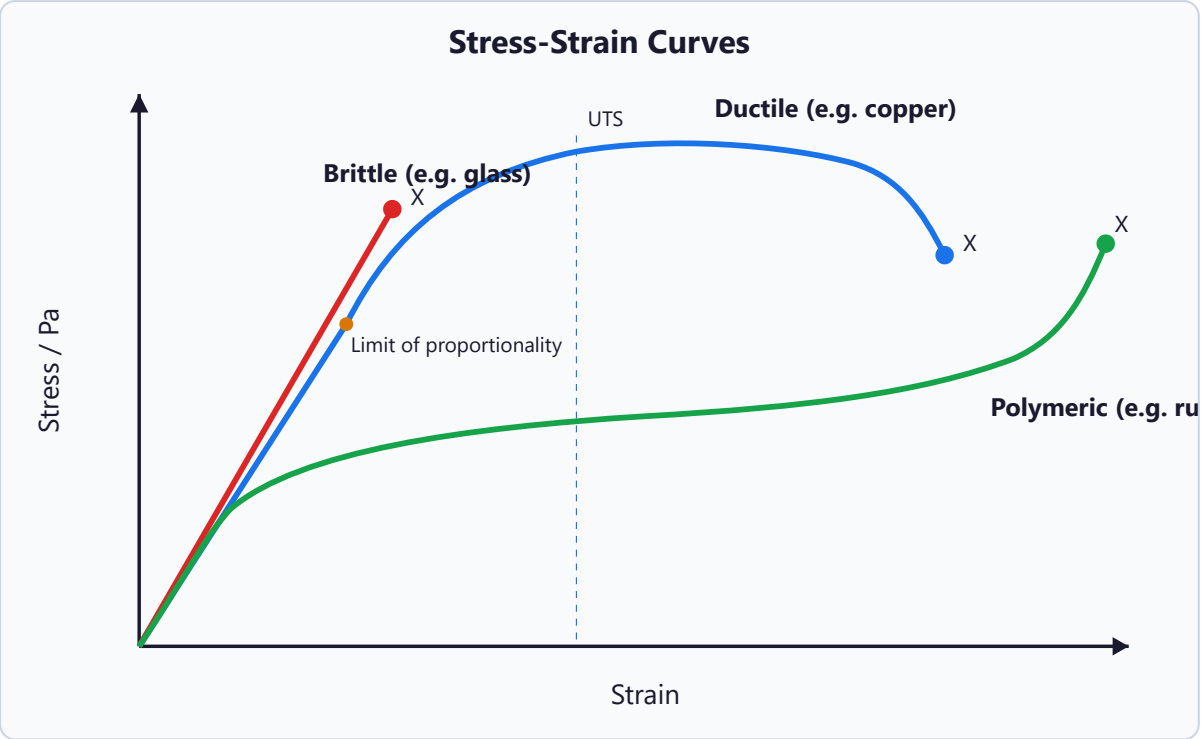
The shape of a stress-strain graph reveals important information about material behaviour:

- **Limit of proportionality:** The point beyond which stress is no longer proportional to strain
- **Elastic limit:** The maximum stress from which the material can return to its original shape when the load is removed
- **Yield point:** The stress at which the material begins to deform plastically (permanent deformation)
- **Ultimate tensile stress (UTS):** The maximum stress the material can withstand before failure
- **Breaking point:** The stress at which the material fractures

Types of Material Behaviour

Behaviour	Description	Examples
Brittle	Breaks with little or no plastic deformation. The stress-strain graph is linear up to the breaking point.	Glass, ceramics, cast iron
Ductile	Can be drawn into wires. Shows significant plastic deformation before fracture.	Copper, gold, mild steel
Polymeric	Can undergo very large strains. Often shows non-linear elastic behaviour with hysteresis.	Rubber, polyethylene

© RAG Learning · Not for redistribution



Elastic and Plastic Deformation

Elastic deformation: Deformation where the material returns to its original shape when the load is removed. The atoms are displaced from their equilibrium positions but return when the force is removed. Energy is stored and fully recovered.

Plastic deformation: Permanent deformation where the material does not return to its original shape. Atoms move to new equilibrium positions. Energy is dissipated (not recovered).

The area under a force-extension graph represents the work done (energy stored or transferred). For elastic deformation, this equals the elastic strain energy. For loading and unloading curves that do not overlap (hysteresis), the area between the curves represents energy dissipated as heat.

$$\text{Elastic strain energy} = \tfrac{1}{2}\,F \times \Delta L = \tfrac{1}{2}\,k(\Delta L)^2$$

3.6 Sensing

Potential Dividers

A potential divider is a circuit with two or more resistors in series across a power supply. The voltage is divided between the resistors in proportion to their resistances.

$$V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

By replacing one resistor with a sensor (such as a thermistor or LDR), the output voltage changes in response to the physical quantity being measured. This is the basis of many sensing circuits.

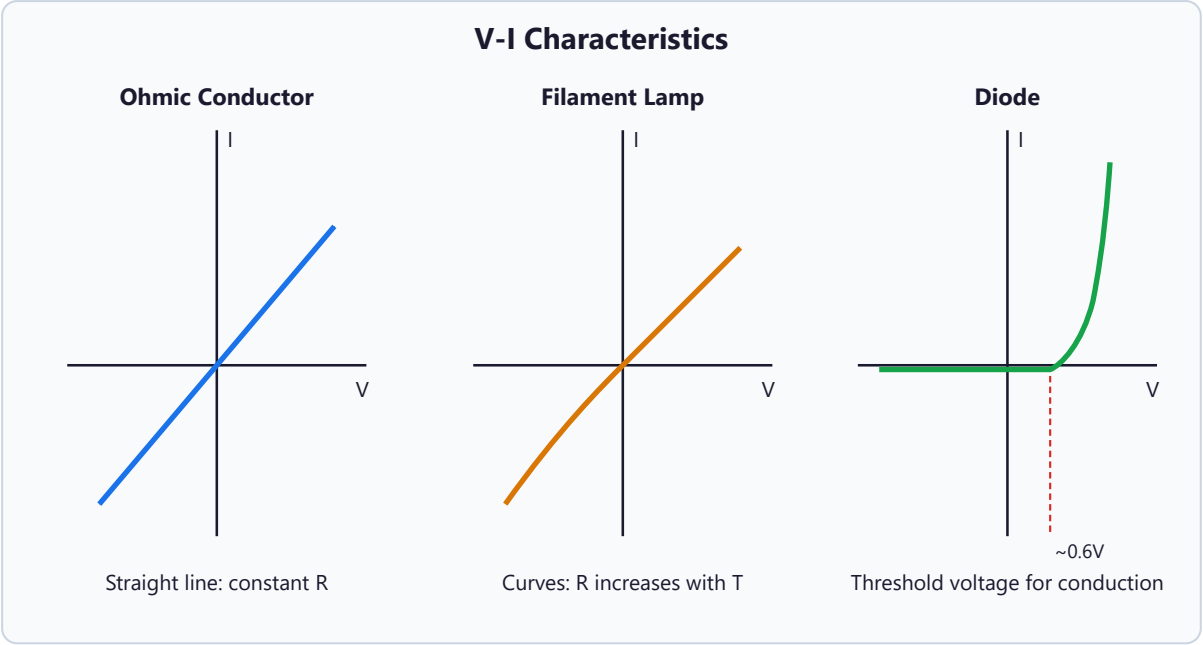
Sensors and Transducers

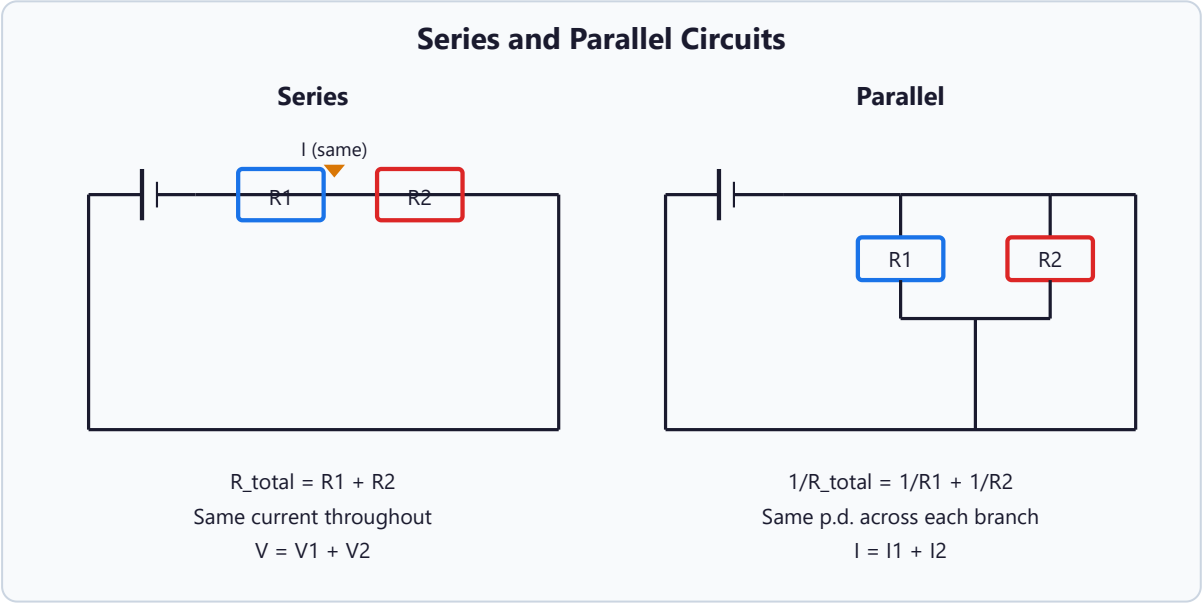
Sensor: A device that detects a physical quantity and produces a signal (usually electrical) in response.

Transducer: A device that converts energy from one form to another. An input transducer converts a physical quantity into an electrical signal; an output transducer converts an electrical signal into another form.

Sensor	Quantity Detected	How It Works
Thermistor (NTC)	Temperature	Resistance decreases as temperature increases
LDR	Light intensity	Resistance decreases as light intensity increases
Strain gauge	Strain / Force	Resistance changes as the gauge is stretched or compressed
Piezoelectric crystal	Pressure / Vibration	Generates a voltage when mechanically stressed

© RAG Learning · Not for redistribution





Signal Conditioning

Raw sensor signals often need processing before they can be used:

- **Amplification:** Increasing the voltage or current of a weak signal
- **Filtering:** Removing unwanted frequency components (noise)
- **Analogue-to-digital conversion (ADC):** Converting a continuous signal to digital for processing by a microcontroller

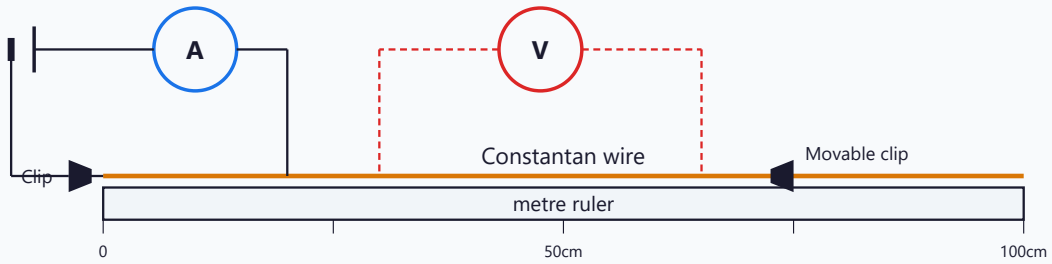
Exam Tip: In sensing circuit questions, always consider the sensitivity (how much the output changes per unit change in input), the response time (how quickly the sensor responds), and the range (the minimum and maximum values the sensor can detect). These determine whether a sensor is suitable for a given application.

Calibration

Sensors must be calibrated against known standards to ensure their readings are accurate. Calibration involves comparing the sensor output with reference values and creating a calibration curve or lookup table. For example, a thermistor-based thermometer would be calibrated using ice-water (0 °C) and boiling water (100 °C) as reference points.

Practical Activity: Resistivity of a Wire

Resistivity of a Wire — Apparatus Setup



Measure wire diameter with micrometer at 3+ points

Apparatus

- Constantan wire (approx 1 m), taped to a metre ruler
- Crocodile clips (one fixed, one movable along the wire)
- DC power supply or battery pack
- Ammeter (in series) and voltmeter (in parallel across the wire section)
- Micrometer screw gauge (to measure wire diameter)
- Switch

Method

1. Measure the diameter of the wire at three different points along its length using the micrometer, rotating the wire 90 degrees each time. Calculate the mean diameter and hence the cross-sectional area: A = pi(d/2)^2.
2. Set up the circuit with the ammeter in series and the voltmeter in parallel across the section of wire being tested.
3. Fix one crocodile clip at the 0 cm mark. Connect the other clip at 10 cm to give a length L = 0.100 m.
4. Close the switch briefly, record the ammeter and voltmeter readings, then open the switch (to avoid heating the wire).
5. Calculate resistance: R = V / I.
6. Repeat for lengths of 20 cm, 30 cm, 40 cm, ..., up to 100 cm.
7. Plot a graph of R (y-axis) against L (x-axis). The gradient equals rho / A, so rho = gradient x A.

Safety

- Keep the current low (below 1 A) and only switch on briefly to prevent the wire heating up (which changes its resistance)
- Do not touch the wire while current is flowing as it may become hot
- Ensure the wire is securely taped to the ruler so it does not spring off

Analysis

Plot R against L. For a uniform wire, R = rhoL / A, so the graph should be a straight line through the origin with gradient = rho / A. Calculate rho = gradient x A. The accepted value for constantan is approximately 4.9 x 10^-7 ohm m.

Sources of Error & Improvements

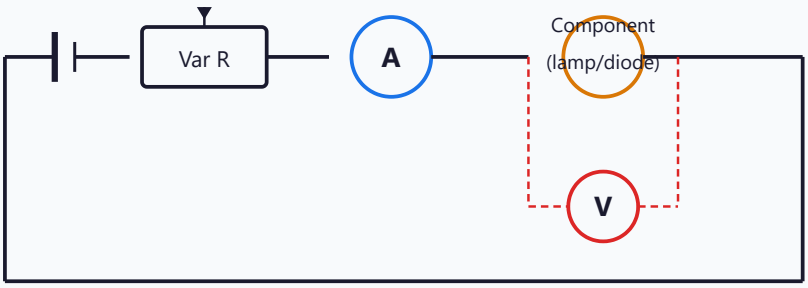
Source of Error	Type	Improvement
Wire heats up during measurement, changing resistance	Systematic	Keep current low, switch on briefly, allow cooling between readings

Difficulty measuring exact length to crocodile clip contact point	Random	Use large range of lengths; clip position uncertainty is small fraction of total
Wire diameter not uniform	Systematic	Measure diameter at multiple points and orientations, use mean
Zero error on micrometer	Systematic	Check zero reading before use and subtract any zero error

Exam Tip: The biggest source of uncertainty is usually the diameter measurement, since it is squared when calculating area. A small percentage error in d leads to double that percentage error in A, and hence in rho.

Practical Activity: V-I Characteristics

V-I Characteristics — Circuit Setup



Vary the variable resistor to change V and I

Apparatus

- DC power supply (or battery pack)
- Variable resistor (rheostat) to vary the p.d. across the component
- Ammeter (in series)
- Voltmeter (in parallel with the component)

- Component under test: filament lamp, diode, or fixed resistor
- Switch and connecting leads

Method

1. Connect the circuit with the ammeter in series and the voltmeter in parallel across the component.
2. For the filament lamp: vary the variable resistor to obtain a range of voltages from 0 V up to the rated voltage. Record V and I at each setting. Reverse the battery connections and repeat to obtain negative V and I values.
3. For the diode: start with the diode in forward bias. Increase voltage gradually from 0 V. Note the threshold voltage (~0.6 V for silicon) where current begins to flow significantly. Reverse the connections (reverse bias) — current should be effectively zero.
4. Plot I (y-axis) against V (x-axis) for each component.

Safety

- Do not exceed the rated voltage of the filament lamp
- The filament lamp may become very hot — do not touch during or immediately after the experiment
- Do not exceed the maximum reverse voltage for the diode (to avoid breakdown)

Analysis

An ohmic conductor gives a straight line through the origin. The filament lamp curve shows increasing gradient (resistance increases with temperature). The diode shows negligible current below the threshold voltage, then a rapid exponential increase in forward bias, and near-zero current in reverse bias.

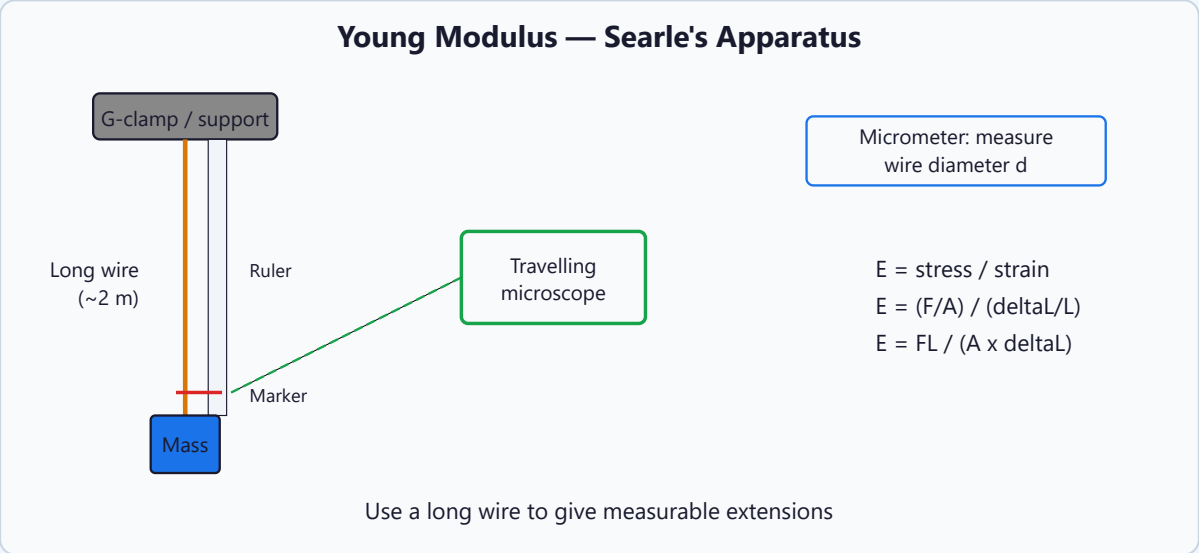
Sources of Error & Improvements

Source of Error	Type	Improvement
Lamp temperature changes during readings, affecting resistance	Systematic	Allow the lamp to reach thermal equilibrium at each voltage before recording

Ammeter has internal resistance, affecting measured values	Systematic	Use a low-resistance ammeter; use a potential divider instead of variable resistor for finer control

Exam Tip: A potential divider circuit allows you to vary the voltage from 0 V to the supply voltage, whereas a simple variable resistor in series cannot reduce the voltage to zero. Use a potential divider for components (like diodes) where you need fine voltage control near zero.

Practical Activity: Determining the Young Modulus



Apparatus

- Long thin copper or steel wire (~2 m), clamped securely at the top
- Metre ruler or reference wire for measuring original length
- Micrometer screw gauge (to measure wire diameter)
- Slotted masses and hanger
- Travelling microscope or Vernier scale (to measure small extensions)
- Marker or tape on the wire to track extension
- Safety goggles

Method

1. Measure the original length L of the wire from the clamp to the marker using a metre ruler.
2. Measure the diameter d of the wire at several points using the micrometer. Calculate mean d, then cross-sectional area A = pi(d/2)^2.
3. Attach the mass hanger. Start with a small initial load to straighten the wire (do not include this in data).
4. Add known masses (e.g., 100 g increments). For each mass, record the extension delta-L using the travelling microscope or Vernier scale.
5. Continue adding masses until you have at least 6 data points. Do not exceed the elastic limit.
6. Unload the masses to check the wire returns to its original length (confirming elastic behaviour).
7. Plot stress (F/A) against strain (delta-L / L). The gradient of the linear region is the Young modulus E.

Safety

- Wear safety goggles — the wire is under tension and could snap
- Place a cushion or sand tray beneath the masses in case they fall
- Do not stand directly beneath the hanging masses

Analysis

The gradient of the stress-strain graph in the linear (Hooke's law) region gives the Young modulus E directly. Alternatively, plot F against delta-L and use E = gradient x L / A.

Sources of Error & Improvements

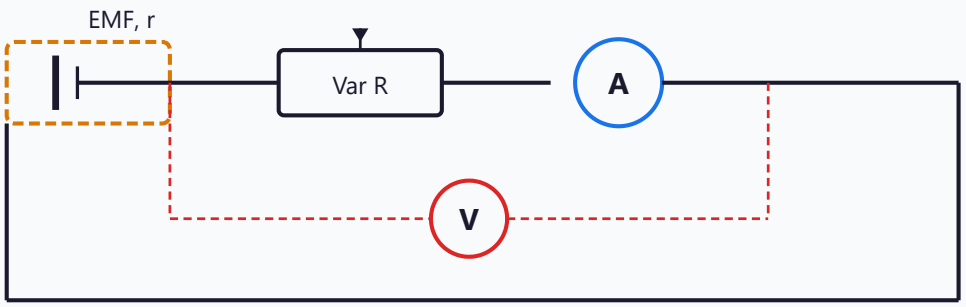
Source of Error	Type	Improvement
Difficulty measuring small extensions accurately	Random	Use a long wire to increase extensions; use a travelling microscope for high precision
Wire diameter not uniform	Systematic	Measure at multiple points and orientations, use the mean

Wire slipping at the clamp	Systematic	Ensure clamp is tight; check by loading and unloading to confirm no slippage
Thermal expansion from temperature changes	Systematic	Use a reference wire alongside (Searle's method) so thermal effects cancel

Exam Tip: The largest percentage uncertainty is almost always in the diameter measurement (it is squared). Using a long thin wire maximises the extension for a given stress, making measurements more accurate.

Practical Activity: EMF and Internal Resistance

EMF and Internal Resistance



V = EMF - Ir, so plot V vs I: gradient = -r, y-intercept = EMF

Apparatus

- Cell or battery (with non-negligible internal resistance)
- Variable resistor (rheostat)
- Ammeter (in series)
- Voltmeter (in parallel across the cell)
- Switch and connecting leads

Method

1. Set up the circuit with the variable resistor, ammeter in series, and voltmeter across the cell terminals.
2. Adjust the variable resistor to its maximum resistance. Close the switch, record V and I.
3. Gradually decrease the resistance, recording V and I at each setting (at least 6 pairs).
4. Open the switch between readings to minimise energy drain on the cell.
5. Plot V (y-axis) against I (x-axis).

Safety

- Do not short-circuit the cell (resistance too low) as it will overheat and may leak
- Switch off between readings to prolong cell life

Analysis

From V = EMF - Ir, the graph of V against I is a straight line with y-intercept = EMF and gradient = -r. The internal resistance r is the magnitude of the gradient. The EMF is the open-circuit voltage (when I = 0).

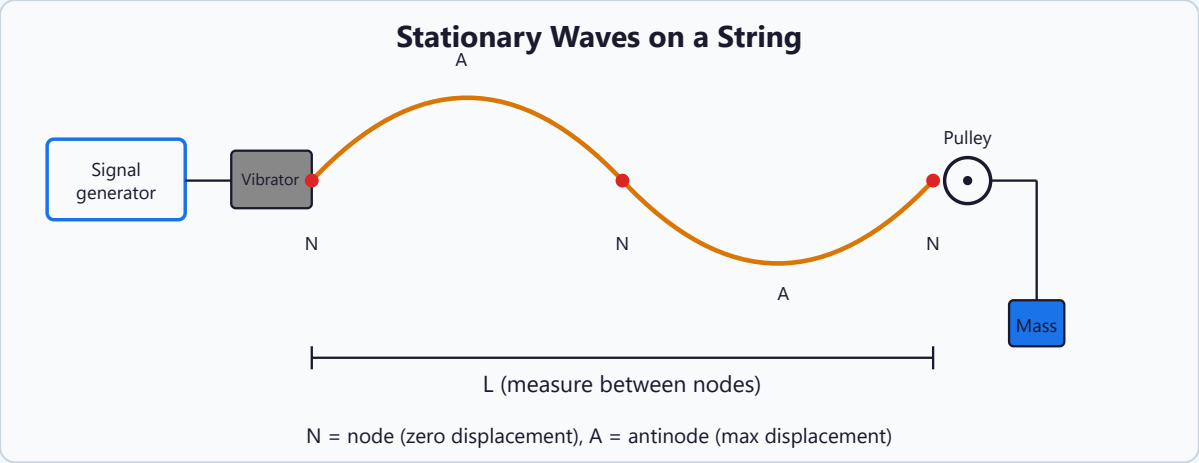
Sources of Error & Improvements

Source of Error	Type	Improvement
Cell EMF decreases over time as it runs down	Systematic	Take readings quickly; switch off between readings; use a fresh cell
Contact resistance at connections	Systematic	Use clean, tight connections; use crocodile clips or solder

Exam Tip: The y-intercept gives the EMF because when I = 0 there is no voltage lost across the internal resistance. You cannot measure EMF directly with a voltmeter connected to a cell in a complete circuit because the voltmeter draws some current.

© RAG Learning · Not for redistribution

Practical Activity: Stationary Waves on a String



Apparatus

- Signal generator connected to a vibration generator (mechanical oscillator)
- String (e.g., nylon), attached to the vibrator at one end
- Pulley clamped at the other end of the bench
- Slotted masses on a hanger (to provide tension in the string)
- Metre ruler (to measure the length between nodes)

Method

1. Set up the apparatus with the string passing over the pulley and the masses hanging from the end.
2. Set a fixed mass (tension) and slowly increase the frequency on the signal generator from a low value.
3. Observe the string for the first harmonic (fundamental): one antinode, two nodes at the ends. Record this frequency f1.
4. Continue increasing frequency to find the 2nd harmonic (two antinodes), 3rd harmonic, etc.
5. Measure the vibrating length L between the nodes at each end.
6. For the fundamental: wavelength = 2L, so wave speed v = f1 x 2L.
7. Investigate how changing the tension (by changing the mass) or the length affects the resonant frequency.

© RAG Learning · Not for redistribution

Safety

- Place a cushion or sand tray under the hanging masses
- Wear safety goggles in case the string snaps
- Keep fingers away from the vibrating string

Analysis

For a string of mass per unit length mu under tension T, the wave speed is v = sqrt(T/mu). The fundamental frequency is f1 = v/(2L) = (1/2L)sqrt(T/mu). Plotting f1 against 1/L (for constant T) should give a straight line through the origin. Or plot f1^2 against T to verify the relationship.

Sources of Error & Improvements

Source of Error	Type	Improvement
Difficulty judging exact position of nodes	Random	Use a card rider (small piece of paper) that stays still at nodes but is thrown off at antinodes
Signal generator frequency not exact	Systematic	Calibrate with a known frequency source or use a digital frequency meter
String sags under its own weight	Systematic	Use a light string with sufficient tension to minimise sag

Exam Tip: At the fundamental frequency, the wavelength is twice the string length (lambda = 2L). For the nth harmonic, lambda = 2L/n. The speed of the wave on the string depends on the tension and the mass per unit length, not on the frequency.

Worked Example: A converging lens has focal length 15 cm. An object is placed 25 cm from the lens. Find the image distance and magnification.

1/v = 1/f − 1/u = 1/15 − 1/25 = (5 − 3)/75 = 2/75
v = 75/2 = **37.5 cm** (positive, so the image is real and on the opposite side)

Magnification m = v/u = 37.5/25 = **1.5** (image is 1.5 times larger than the object, and inverted)

Worked Example: A diffraction grating has 500 lines per mm. Calculate the angle of the second-order maximum for light of wavelength 550 nm.

$d = 1/500$ mm $= 1/(500 \times 10^3)$ m $= 2.0 \times 10^{-6}$ m

$d \sin\theta = n\lambda$, so $\sin\theta = n\lambda/d = 2 \times 550 \times 10^{-9} / 2.0 \times 10^{-6} = 0.55$

$\theta = \sin^{-1}(0.55) =$ **33.4°**